



Ocular Plaque Brachytherapy for Uveal Melanoma and Retinoblastoma

19

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Abstract

The main topic of this chapter is brachytherapy for ocular tumor. A small dish-shaped applicator is temporarily attached outside of the eyeball where the tumor is located, and after adequate irradiation time, it is removed. Because generally eyeball moves quickly and frequently, it is difficult to irradiate intraocular tumor with external beam technique. When big dose is delivered with adequate margin to compensate the organ motion, it is difficult to preserve visual function. On the other hand, when the applicator is accurately attached to the location of the tumor, the applicator moves together with the eyeball, and it is possible to perform very conformal irradiation. Therefore, ocular plaque brachytherapy is very important for the management of ocular tumors such as uveal melanoma and retinoblastoma.

Keywords

Plaque brachytherapy · Ocular tumors · Uveal melanoma · Retinoblastoma

19.1 Uveal Melanoma

19.1.1 Introduction

Uveal melanoma is the most common type of primary ocular melanoma in adult, accounting for approximately 79–81% of ocular melanoma cases [1]. Similar to skin melanoma, incidence of uveal melanoma has racial difference: incidence of

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veal melanoma was reported to be 6.02 cases in 1 million population in non-Hispanic white [2] while 0.25 cases in 1 million population in Japanese [3]. Although local control of the tumor is generally excellent, because up to 20–50% of patients will eventually develop distant metastasis, its mortality is relatively high. Because the eyeball has no regional lymph node, typical mode of metastasis is hematogenous metastasis, and the liver is the most commonly affected organ (89%) followed by the lung (29%) and bone (17%) [4]. UV-B radiation exposure and wearing no sunglasses in the daylight have reported to be associated with the development of uveal melanoma [5].

Typically uveal melanoma patients present with visual symptoms such as visual field defect, floaters in vision, photopsia, and blurred vision. On the other hand, approximately 30% of patients are asymptomatic and are diagnosed by chance. Different from other malignancies, diagnosis of uveal melanoma may often be made by no pathological confirmation because of the possibility of developing extraocular dissemination, intraocular bleeding, or retinal detachment through biopsy procedure. Most diagnoses are made by normal funduscopic examination with or without ultrasonography and fluorescein angiography. The major differential diagnosis for uveal tumors would be metastasis from other sites of origin and pigmented nevi. It is difficult to distinguish small-sized melanoma from pigmented nevi, and Shields et al. reported that possibility of progressing malignant melanoma from pigmented nevi was 2%, 9%, and 13% at 1, 5, and 10 years of follow-up, respectively [6]. Therefore, pigmented nevi which look similar to melanoma need to be closely followed for potential malignant transformation.

Risk factors which relate to poor prognosis of uveal melanoma include advanced age at presentation, male gender, large tumor size, extraocular extension, involvement of the ciliary body, and uncontrolled tumor after initial local treatment. One of the most important prognostic factors of uveal melanoma is tumor size (largest basal diameter (LBD) and thickness). The Collaborative Ocular Melanoma Study (COMS) group created staging system according to tumor size (Table 19.1) and conducted a series of prospective clinical trials according to the tumor categories [7–10]. The 5-, 10-, and 12-year mortalities of medium-sized tumor were 10%, 18%, and 21%, respectively, for patients treated by I-125 plaque brachytherapy and 11%, 17%, and 17% for patients treated by enucleation, showing no inferiority of plaque brachytherapy [7]. In the COMS large tumor trial, disease-associated mortality at 5 and 10 years was 28% and 40% for patients treated by enucleation [8]. Monosomy of chromosome 3 in the tumor cell is reported to be prone to distant metastasis and associated with a worse prognosis [11]. BAP1 (BRCA1-associated protein 1 gene) mutations are reported to be present in about 1.6% of uveal melanoma patients and are also associated with

Table 19.1 The Collaborative Ocular Melanoma Study (COMS) staging of melanoma of the eye

Small	Between 1 mm and 3 mm in height and between 5 mm and 16 mm LBD
Medium	Between 3.1 mm and 8 mm in height and no more than 16 mm LBD
Large	More than 8 mm in height or more than 16 mm LBD

LBD: largest basal diameter

larger tumor diameters, more frequently ciliary body involvement, and higher rate of distant metastasis resulting in poor prognosis [12]. However, if the tumor is localized, no biopsy will be performed due to the possibility of development of iatrogenic dissemination as written above, and no treatment strategy based on tumor biology is currently performed for localized disease in daily practice. On the other hand, in the United States or European countries, because of high prevalence of this disease and existence of well-experienced ocular pathologists, needle biopsy is incorporated into diagnostic work-up in dedicated institutions using very thin needles (25–27 G needles). Because no adequate tissue specimen can be obtained by needle biopsies, to compensate its weakness, there is a primary tumor gene expression profile (GEP) testing which can categorize patients into two groups: low-risk Class 1 group with low metastatic potential and high-risk Class 2 group [13]. In terms of local management of uveal melanoma, there is no difference between these two groups. After local treatment, high-risk Class 2 group patients could be monitored frequently for distant metastasis and can be offered a clinical trial trying to prevent development of distant metastasis.

19.1.2 Diagnosis

As described above, normally biopsy is avoided due to the possibility of developing extraocular dissemination, intraocular bleeding, and retinal detachment; patient history and physical examination taken by conventional slit lamp and ophthalmoscopy are very important for the diagnosis of uveal melanoma. High accuracy of diagnosis without pathological confirmation was demonstrated by one of COMS [10]. In addition to these, photography, fundus autofluorescence imaging, MRI, and I-123 IMP SPECT (123N-isopropyl-p-[123I]-iodoamphetamine single photon emission computed tomography) are also used in the diagnostic procedure. Although I-123 IMP SPECT has high sensitivity and specificity, it is not popular in the United States and Europe because they perform needle biopsy and GEP as described above. When it is difficult to make diagnosis, especially for small tumors, treatment will be started after confirmation of disease progression.

19.1.3 Local Treatment for Uveal Melanoma

Treatment of the primary tumor depends upon tumor size, location, general condition of the patient, and patient preference. Historically, enucleation had been the treatment of choice. Since equal effectiveness of local control and overall survival between surgical resection and plaque brachytherapy with I-125 were demonstrated for medium-sized uveal melanoma (<8 mm in height and <16 mm in diameter) by COMS phase III clinical trial [7], the majority of medium-sized localized disease are treated with plaque brachytherapy to spare vision in the United States. On the contrary, only one institution, the National Cancer Center Hospital, can perform plaque brachytherapy using Ru-106 in Japan, and the number of patients

who can be treated within a certain period of time by Ru-106 is strictly limited by the law; majority of patients treated by plaque brachytherapy are retinoblastoma patients, and unfortunately patients who are good candidate for plaque brachytherapy are usually treated with external beam radiation therapy such as stereotactic radiation therapy or particle beam with which it is difficult to preserve adequate visual function. I-125, which can treat thicker tumor than Ru-106, is not allowed to be used for ocular plaque brachytherapy in Japan; therefore, urgent introduction of I-125 in the management of ocular tumor is warranted in Japan. Exclusion criteria for plaque brachytherapy described in the American Brachytherapy Society consensus guideline include tumors with extraocular extension more than 5 mm in diameter (T4e), blind painful eyes, and patients with no light perception vision [14]. Enucleation is still the treatment of option in selected cases when there is little probability of preserving eyesight, the tumor is too large to be treated by other local therapies, a tumor causes extensive bleeding necessitating urgent resection, or the patients feel anxious for eye-preservation treatment. The COMS quality-of-life study found that although patients treated with brachytherapy enjoyed significantly better visual function than patients treated with enucleation, patients treated with brachytherapy were more likely to have anxiety during follow-up than patients treated with enucleation [15].

Twelve-year results of COMS phase III clinical trial for medium-sized tumor of I-125 brachytherapy vs enucleation found that cumulative all-cause mortality rates and death with histopathologically confirmed melanoma metastasis were 43% and 21% in I-125 brachytherapy arm and 41% and 17% in enucleation arm, respectively [7]. Among brachytherapy patients, the most common complications during the first 5 years after treatment were loss of visual acuity and local recurrence [16]. Five-year visual acuity of 20/200 or worse was 63%, and 5-year cumulative rate of enucleation was 12%. Long-term results of carbon ion radiation therapy for locally advanced (UICC T2–T3) uveal melanoma were reported, and 5-year overall survival and local control were 80.4% and 92.8% with 35.9% of neovascular glaucoma incidence rate [17].

If the tumor is small enough to be treated with plaque brachytherapy, brachytherapy should be considered in terms of both local control and organ function preservation.

19.1.4 Treatment-Related Complications

Radiation maculopathy or radiation optic neuropathy could be occurred especially posteriorly located tumors. These late complications related with plaque brachytherapy generally occur within 3 years after treatment. Toxic tumor syndrome can occur after large melanomas treated by radiotherapy either with plaque brachytherapy or proton beam radiation by inducing cytokines and cause ischemia of the retina and exudative retinal detachment. This radiation maculopathy is the primary cause of irreversible visual loss. Even if the tumor is controlled, enucleation is needed when toxic tumor syndrome is developed.

19.2 Retinoblastoma

19.2.1 Introduction

Retinoblastoma is a disease for neonatal infant or baby and is the most frequent intraocular malignancy in children. There is no difference of incidence according to race or sex. In Japan, about 70–90 patients are diagnosed with retinoblastoma every year [18], and 95% of them are diagnosed until they become 5 years old. About 60% of the patients are non-hereditary and unilateral, 25% are hereditary and bilateral, and 15% are hereditary and unilateral. Patients with the genetic diathesis are prone to develop secondary malignancies, especially soft tissue sarcomas and osteosarcomas.

19.2.2 Diagnosis

Typical symptom for retinoblastoma for sporadic cases is leukocoria or strabismus. Because infants cannot complain about visual disturbance by themselves, most cases are found in locally advanced stage. When relatives have family history of retinoblastoma, early fundus examination occasionally finds early-stage disease. Same as uveal melanoma, because of concern for developing dissemination, needle biopsy is thought to be a contraindication. Diagnosis is generally done by ocular examination, detecting whitish tumors with typical calcification. Differential diagnoses are persistent hyperplastic primary vitreous (PHPV), Coats disease, retinal dysplasia, ocular toxocariasis, congenital cataract, and retinal detachment. Reese-Ellsworth classification had been used to predict prognosis when external beam radiation therapy was the main treatment modality, but now that chemotherapy became the main player for retinoblastoma [19], it does not correlate well with prognosis. The International Classification of Retinoblastoma (ICRB) categorizes (Table 19.2) intraocular tumor very well and correlates with clinical results [20]. Since the 7th edition of UICC TNM, the concept of ICRB category has been adhered fundamentally; therefore the succeeding 8th edition of TNM staging system [21] also correlates well with prognosis of retinoblastoma.

Table 19.2 The International Classification of Retinoblastoma (ICRB)

Group A	Small intraretinal tumors (<3 mm) away from fovea and disc
Group B	Tumors >3 mm, macular or juxtapapillary location, or with subretinal fluid
Group C	Tumor with focal subretinal or vitreous seeding within 3 mm of tumor
Group D	Tumor with diffuse subretinal or vitreous seeding >3 mm from tumor
Group E	Extensive retinoblastoma occupying >50% of the globe with or without neovascular glaucoma, hemorrhage, extension of tumor to optic nerve or anterior chamber

19.2.3 Treatment

When tumor is localized in the retina, ocular preservation probability is very high, and the ultimate goal of the treatment is visual preservation. Localized tumor thickness of which is less than 3 mm can be well controlled by laser or cryotherapy, and approximately 90% local control will be expected. Moderately large tumor thickness of which is less than 5 mm can be well controlled by Ru-106 brachytherapy and less than 8 mm by I-125. Tumors thicker than 5 mm or tumors having retinal or vitreous dissemination require induction chemotherapy followed by local therapy, and organ preservation rate for such tumors is about 50% with poor visual function. There exists controversy whether to preserve the eyeball which has no adequate visual function. Even without visual function, if the eyeball is preserved, the orbital growth will be encouraged, and better cosmetic result will be expected. Even though tumor is found in locally advanced stage, only if the tumor is localized within the eye, 95% of patients can expect long-term survival. On the other hand, when tumor develops extraocular spread, prognosis becomes very poor and requires multidisciplinary modalities to save the life.

19.2.4 Tumors with Less Than 3 mm Thickness (TNM(7th) T1, TNM(8th) T1)

Either laser or cryotherapy can control such small tumors. The tumors located in peripheral part of the retina are good candidate for periscleral cryotherapy. Both techniques can achieve approximately 90% of local control.

19.2.5 Tumors with Less Than 5 mm Thickness or Limited Vitreous Dissemination (TNM(7th) T2a, TNM(8th) T2a)

This category of tumors is good candidate for plaque brachytherapy, and 94% of 5-year tumor control is reported by Ru-106 plaque brachytherapy [22].

19.2.6 Tumors Larger Than 5 mm and/or Widespread Dissemination (TNM(7th) T2b–T3, TNM(8th) T2b)

Several decades ago, external beam radiation therapy (EBRT) had been given for such locally advanced patients. However, because of the possibility of development of secondary malignancies, especially in patients with hereditary disease, and concern for facial bone growth obstruction, chemotherapy has replaced the position of EBRT [20]. Typical agents used as systemic chemotherapy are combination of vincristine, carboplatin, and etoposide. After achievement of optimal tumor shrinkage, which is so-called chemoreduction, local therapies described above are applied because chemotherapy alone is not strong enough to completely destroy the tumor. To reduce the toxicity associated with systemic chemotherapy such as

myelosuppression, infection, nausea and vomiting, renal dysfunction, ototoxicity, reproductive organ damage, and secondary malignancies, selective ophthalmic arterial injection (SOAI) was introduced [23]. Suzuki et al. reported long-term clinical result of retinoblastoma patients treated by SOAI using melphalan [24]. Eyes with active vitreous seeding >3 mm from tumor are also treated with vitreous injection of melphalan. Although plaque brachytherapy is not suitable for such large and disseminated tumor, combined with systemic chemotherapy, SOAI, or vitreous injection of chemotherapy and if favorable tumor shrinkage is obtained, plaque brachytherapy is also applied [25].

19.2.7 Tumor with Glaucoma, Massive Intraocular Bleeding, or Orbital Inflammatory Disease (TNM(7th) T3b, TNM(8th) T3)

Primarily eyes with such complications should be enucleated. If pathologic report revealed the risk factor of metastasis such as massive choroidal involvement, optic nerve invasion, or extrascleral spread, adjuvant chemotherapy will be given to prevent hematogenous metastasis, central nervous system metastasis, or orbital recurrence, respectively. Chemotherapy agents given are similar to those used in treatment for neuroblastoma or regimen according to those used for limited stage Retinoblastoma such as VEC (vincristine, etoposide, and carboplatin).

19.2.8 Complications

Radiation retinopathy/maculopathy and optic neuropathy represent late-onset complications of plaque brachytherapy and are characterized by a slowly progressing occlusive vasculopathy, which produces variable ischemic changes and subsequent visual loss [26]. Other types of complications include exudative retinal detachment, radiation cataract, neovascular glaucoma, and retinal bleeding.

19.3 Plaque Brachytherapy Procedure

Application of brachytherapy in the management of intraocular tumor has been started since 1930 [27]. Various types of isotopes have been used as plaque sources including Co-60, Ru-106, I-125, Cs-131, Sr-90, and Pd-103 [17]. Ru-106 emits beta ray and, therefore, can only treat thin tumors less than 5–6 mm in apical thickness. In contrast, I-125 emits gamma ray and can treat up to 8-mm-thick tumors. Because dose distribution created by Ru-106 is more conformal than I-125, when the tumor apical dose is the same, radiation dose to the contralateral side of the eyeball is far less in Ru-106 compared to I-125 (Fig. 19.1). Ru-106 plaque applicators (BEBIG Isotopen und Medizintechnik GmbH, Berlin, Germany) stocked in the National Cancer Center Hospital are shown in Fig. 19.2. Round shaped applicators, so-called CCA and CCB, are used to treat tumors away

Fig. 19.1 Comparison of dose distribution for Ru-106 and I-125 plaque brachytherapy. The round represents the eyeball and the brown object at the bottom of the eyeball represents a tumor. Dose is normalized to the apex of the tumor (40 Gy). While contralateral side dose with Ru-106 is 0 Gy, it is 3 Gy in I-125

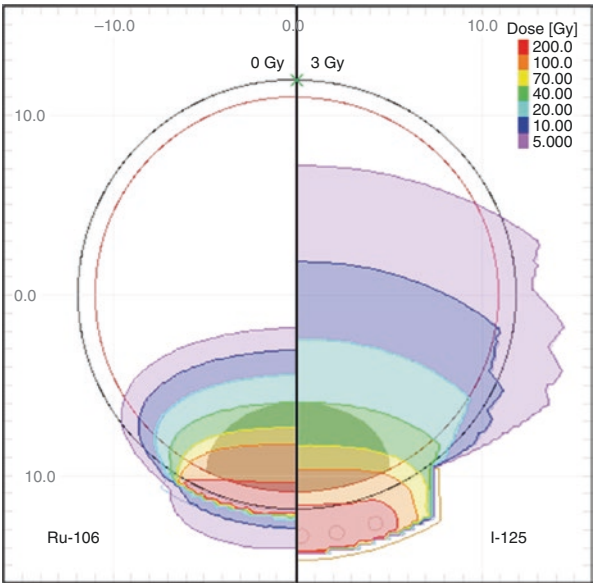
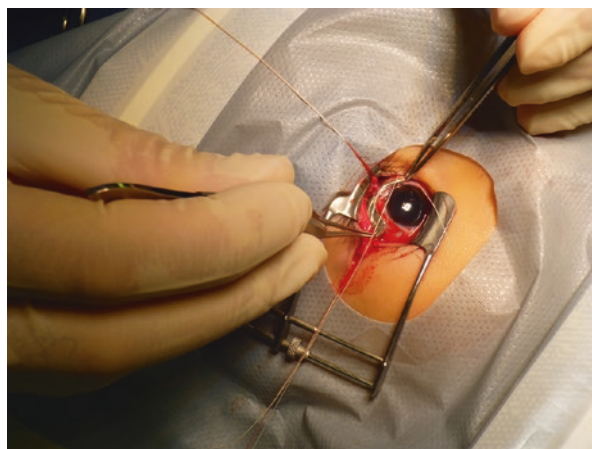


Fig. 19.2 Five different types of applicator are shown. Small applicators are used for small tumors and large applicators for large tumors. The notched applicators are used for anteriorly located tumor to avoid cornea and lens as shown in Fig. 19.3. Applicator, which has deep notch, COC, is used to treat juxtapapillary tumor. Because of the presence of the optic nerve, which connects the eyeball to the brain, it is impossible to put a plaque applicator just behind of a tumor which locates just next to the optic nerve. Therefore, by using such an applicator which has deep notch surrounding the optic nerve, the β rays emitted by Ru-106 turn around, and it can be expected to secure dose of the surface of the optic nerve

from normal structures such as optic nerve, lens, or cornea. Applicators which have shallow notch, CIA and CIB, are used to treat periphery located tumors to avoid high dose to the lens and cornea. Applicator which has deep notch, COC, is used to treat juxtapapillary tumor to ensure the dose which locates very close to

Fig. 19.3 After conjunctival incision, the external ocular muscle is temporarily detached to make applicator as close to the tumor as possible. Applicator is sutured to the eyeball by the surgical thread through eyelets



the optic nerve. Because of the presence of the optic nerve, which connects the eyeball to the brain, it is impossible to put a plaque applicator just behind of a tumor which locates just next to the optic nerve. Therefore, by using an applicator which has deep notch surrounding the optic nerve, the β rays emitted by Ru-106 will turn around, and it can be expected to secure dose of the surface of the optic nerve. Tumor localization is determined by indirect ophthalmoscopy. After conjunctival incision external ocular muscle is temporarily detached to make applicator as close to the tumor as possible. Tumor margins are marked on the sclera by indentation of the sclera using illuminator, and applicator is inserted (Fig. 19.3). Applicator is sutured to the eyeball by the surgical thread through eyelets. Because the thickness of Ru-106 plaque applicator is only 1 mm, it is suitable for children. Prescription dose at the apex of the tumor is 85 Gy and 40 Gy for uveal melanoma and retinoblastoma, respectively.

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